

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

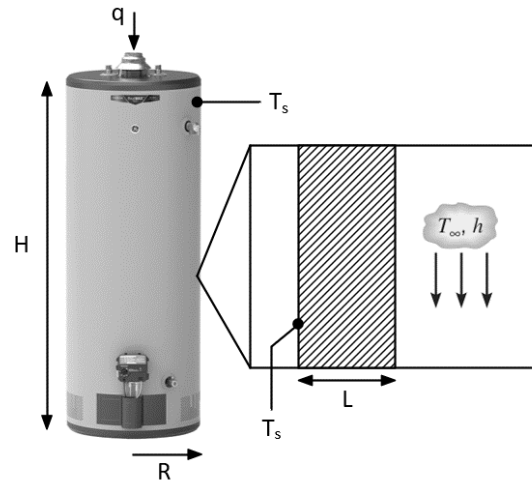
Spring 2026

Problem Set #2

Posted: Monday, Feb 2

Due: Tuesday, Feb 10, 11:59 pm

1. (40 pt) Consider an electric water heater operated under a steady state. This water heater is a cylinder with radius $R = 0.5$ m and height $H = 1$ m. When the water heater is fully filled, the water temperature inside can be maintained at 50°C . The wall of the water heater is thin enough and of high thermal conductivity, so we can assume that the exterior wall temperature of the water heater T_s is also 50°C at the steady state. The water heater is placed in an environment with ambient temperature of $T_\infty = 20^\circ\text{C}$ and heat transfer coefficient of $h = 15$ W/(m²·K). The bottom of the water heater is well thermally insulated. Thermal radiation is negligible in this problem.



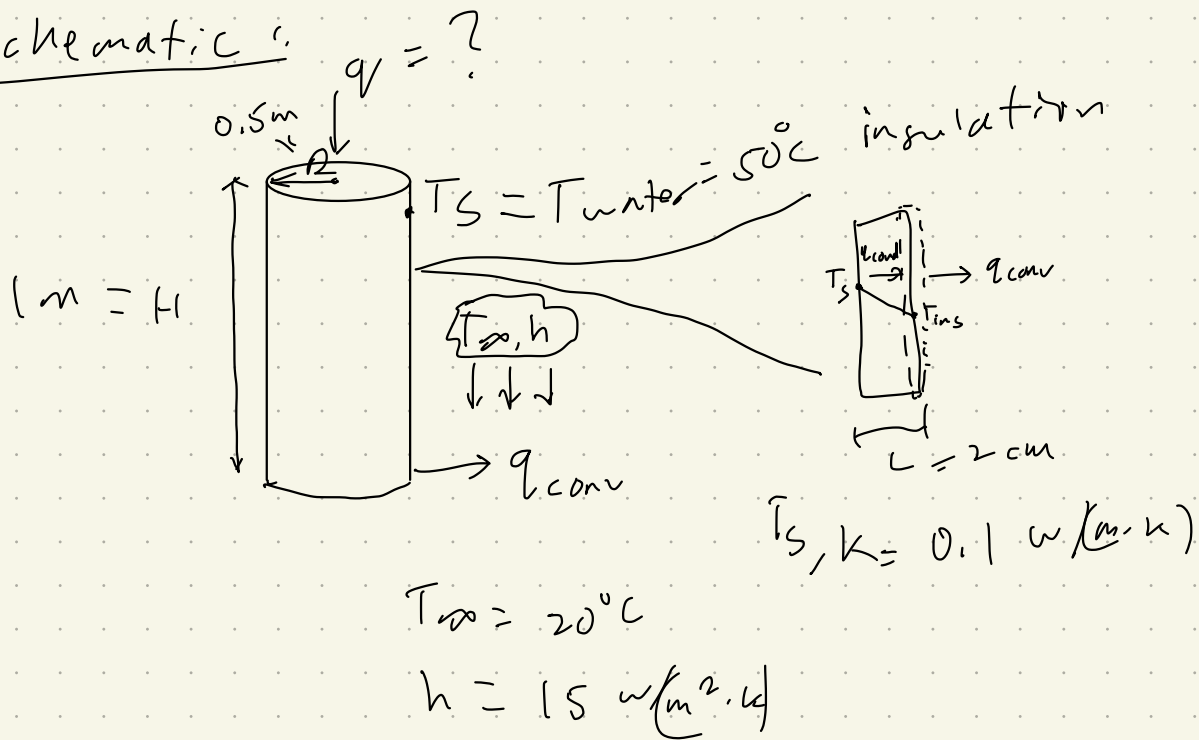
- (a) Calculate the total electrical heating power q needed to maintain a constant water temperature at 50°C .
- (b) To reduce the heat loss from the water heater, we now attach a $L = 2$ cm thick thermal insulation foam to the exterior wall of the water heater, including both side wall and top wall (zoom-in illustration). Thermal conductivity k of the insulation foam is 0.1 W/(m·K). Since the thickness of the insulation foam is much smaller than the radius of the water heater, we can assume 1D heat conduction across the insulation foam. To maintain a constant water temperature at 50°C , now how much total electrical heating power q is needed?
- (c) Compare the results obtained from (a) and (b), discuss the role of thermal insulation in the energy saving of water heater.

1)

Known: $R, H, T_{\text{water}}, T_s, T_{\infty}, h, L_{\text{foam}}, k_{\text{foam}}$

- Find:
- Electrical power to keep $T_{\text{water}} = 50^\circ\text{C}$.
(cancel out convective heat loss)
 - Electrical power to keep $T_{\text{water}} = 50^\circ\text{C}$ after add foam insulation to outside of tank.
 - discussion of results and role of insulation

Schematic:



Assumptions: Steady state, no radiation, bottom is well insulated.

Analysis:

$$a) \quad q_{\text{conv}} = q = hA(T_s - T_\infty)$$

$$A = 2\pi RH + \pi R^2$$

$$q = h(2\pi RH + \pi R^2)(T_s - T_\infty)$$

$$= \left(15 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}\right) \left(2\pi(0.5\text{m})(1\text{m}) + \pi(0.5\text{m})^2\right) (50^\circ\text{C} - 20^\circ\text{C})$$

$$q = 1767.15 \text{ W}$$

b) $E_{\text{in}} = E_{\text{out}}$ for control surface on outside edge of insulation.

$$q_{\text{cond}} = q_{\text{conv}}$$

$$k \frac{T_s - T_{\text{ins}}}{L} = h A (T_{\text{ins}} - T_\infty)$$

$$\frac{0.1}{0.02} (50 - T_{\text{ins}}) = 15 (T_{\text{ins}} - 20)$$

$$50 - T_{\text{ins}} = 3(T_{\text{ins}} - 20)$$

$$T_{\text{ins}} = 27.5^\circ\text{C}$$

$$q_{\text{conv}} = q = h A (T_{\text{ins}} - T_{\infty})$$

$$= \left(15 \frac{\text{W}}{\text{m}^2 \text{K}} \right) \left(2\pi (0.5 \text{ m})(1 \text{ m}) + \pi (0.5 \text{ m})^2 \right) (27.5^\circ \text{C} - 20^\circ \text{C})$$

$$q = 441.79 \text{ W}$$

c)

Much less electrical power is required to maintain the water temperature with the insulation. Thermal insulation saves a lot of energy because less heat transfer leaving the system allows less power required to heat it.